

SONG PEI-LING

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I am a Master's student in ITBE@TUM, focusing on advancing AEC digitalization via Generative AI and Semantic BIM. My expertise spans LLM Agents, RAG systems, and robotic Scan-to-BIM automation, utilizing Python and C# to build intelligent automation tools.

EDUCATION

Technical University of Munich

M.Sc. Information Technologies for the Built Environment

Munich, Germany

Oct 2024 - Present

- Ministry of Education Government Scholarship to Study Abroad (GSSA)
- M.Sc. Thesis: Semi-Synthetic Dataset of Building Energy Archetypes via Visual-RAG (Ongoing)
Focus: Multimodal Large Language Models (MLLM), Vision-Language Models (VLM), Visual-RAG, Topology-Guided Chain-of-Thought, Segment Anything Model (SAM)

Chung Yuan Christian University

B.Arch. Architecture (GPA: 3.77 / 4.0)

Taoyuan, Taiwan

Sep 2018 - Jun 2023

- Excellent Performance in Professional Studies

PROFESSIONAL EXPERIENCE

ILLOCA

San Francisco, CA (Remote)

BIM Data Specialist (Contract)

Nov 2025 - Present

- Constructed ground-truth BIM datasets for proprietary Generative AI models by standardizing topological relationships within Revit to ensure semantic validity for latent space alignment
- Developed data processing workflows to ensure architectural metadata is optimized for AI model training.

SUNHOU Architects & Partners Association

Taoyuan, Taiwan

Research & Development

Apr 2025 - Present

- Architected an AI-driven ERP ecosystem by integrating SQL-based RDBMS with a proprietary Retrieval-Augmented Generation (RAG) system to establish a data-driven Decision Support System.

Working Student

Sep 2020 - Jun 2022

- Coordinated a golf club renovation project (GFA 7,400 m^2), managing construction sequencing, detailed design, budget allocation, quality control, and MEP integration.
- Utilized parametric design (Grasshopper) for complex geometric modeling and re-planning of large-scale campus structures.

RESEARCH EXPERIENCE

Context-Aware Digital Twin Driven by Hybrid RAG

2025

- Architected an orchestration-based RAG system in Python integrating Neo4j (Cypher), PostGIS, and Vector Stores to enable context-aware QA for **Digital Twins**.
- Implemented intelligent query routing using **LangChain** and **LLMs** to decompose complex architectural queries and dispatch them to specialized BIM, GIS, and Design Agents.
- Engineered a full-stack prototype utilizing FastAPI for the agentic backend and CesiumJS/Next.js for 3D geospatial visualization of query results.

Real-Time Building Digital Model Reconstruction Using a Robotic Agent

2025

- Deployed a real-time **ROS 2 SLAM** pipeline on a Unitree Go2 quadruped robot equipped with LiDAR and an onboard mini-PC, enabling autonomous navigation and data acquisition.
- Implemented a Deep Learning framework for 3D semantic segmentation utilizing **Point Transformer V3 (PTv3)** trained on the ScanNet-20 dataset.
- Engineered an automated pipeline combining **RANSAC** for plane fitting, **DBSCAN** for noise filtering, and **Manhattan Alignment** to reconstruct IFC-compliant semantic models from raw sensor inputs.

Agentic Framework for Automated Construction Tendering

2025

- Developed an LLM-driven agent with RAG architecture to parse unstructured BoQ PDFs, utilizing DIN 276 standards for semantic trade classification and autonomous vendor matching.

Hierarchical Tree Species Classification via Satellite Time-Series

2025

- Engineered a **machine learning pipeline** for tree species classification, integrating Sentinel-2 multi-spectral time-series data with extensive feature engineering and hyperparameter optimization.
- Benchmarked **Transformers**, **CNNs**, and **RNNs** against **XGBoost** and Random Forest, implementing hierarchical label structures (leaf type, genus, species) to improve model performance and interpretability.

Indoor Ambient Monitoring

2025

- Engineered an **IoT-based monitoring system** integrating multi-modal sensors (CO, PM2.5, Temperature) via LoRaWAN and OGC SensorThings API for real-time indoor analytics.
- Implemented a Digital Twin pipeline by converting CityGML LOD 3.0 models into 3D Tiles via FME, merging semantic building data with live environmental streams in MongoDB.
- Developed an interactive web dashboard using CesiumJS to visualize **spatiotemporal dynamics**, successfully validating system fidelity against real-world environmental events (e.g., rainfall and occupancy surges).

Interactive Visualization on Geothermal Energy

2025

- Showcased at the 1E9 Festival der Zukunft 2025 (Munich) in the Dome Theater.
- Led the visual production using Blender to create a cinematic representation of a geothermal-powered future city. Utilized volumetric lighting and physics simulations to visualize invisible energy flows.

Constraint Design in Timber Modular Process

2024

- Prototyped a parametric design tool using PyRevit to enforce modular constraints and ensure logically consistent construction sequencing, in collaboration with Drees & Sommer.

CYCU Architecture Digital Research Community Director

2020-2023

- Directed digital fabrication lab operations and mentored 200+ students in Python-based CAD/CAM workflows.
- Organized interdisciplinary workshops bridging computational geometry and robotic fabrication.

SKILLS & PROFICIENCY

Programming: Python: PyTorch, LangChain, FastAPI (Advanced), TypeScript: Three.js (Proficient), C# (Proficient), SQL (Intermediate)

AEC & Geospatial Informatics: Semantic Modeling (IFC, CityGML), Rhino/Grasshopper (Advanced), FME, ArcGIS Pro, PostGIS, Neo4j

Robotics & Vision: ROS 2 (Navigation/SLAM), Computer Vision, LLM Agents, Multi-modal RAG

Development Environment: Docker, Git/CI, Linux (Ubuntu/Robot Env), Visualization (Next.js, React)

Languages: English (Professional Proficiency), German (Basic), Mandarin (Native)